Quick Start Guide
Single Photon SPAD Detector

SPAD-050-C-B-FC
SPAD-050-C-R-FC



1 About This Guide

This document is a condensed quick-start guide intended to support basic installation and operation of the device. This guide does **not** replace the full product manual. The complete operation manual can be downloaded at:

https://www.flimlabs.com/products/single-photon-detector/

Scan the QR code to open the full manual.

> Reading the complete manual is strongly recommended before operating the device, unless the user is already familiar with similar laboratory instrumentation.

2 Safety Information

> Do not operate the device if it is damaged or if safe operating conditions cannot be ensured.

- Read the full manual before installation and use.
- Operate only within the specified electrical and environmental limits.
- Do not open the enclosure.
- Disconnect power before modifying connections.
- Use only undamaged and properly rated cables.

3 Intended Use

The Single Photon SPAD Detector is intended as a compact single-photon detection module for converting incident optical radiation into digital photon event signals by professional users in research laboratories, universities, or industrial environments. It may be used as a stand-alone detector unit or integrated into custom optical and time-resolved measurement systems.

The device must be operated only by qualified personnel familiar with laboratory optical and electronic instrumentation.

4 Product Overview

The Single Photon SPAD Detector is a compact single-photon detector based on an integrated SPAD (Single Photon Avalanche Diode) sensor. Incident photons coupled through an optical fiber are detected and converted into standardized digital output pulses for direct integration with data acquisition systems.

The device is available in two variants with different spectral response ranges and includes integrated thermoelectric cooling (TEC) to stabilize detector performance and reduce dark count rate.

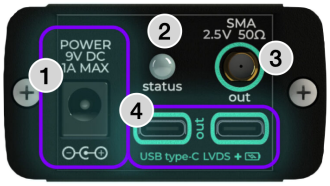
Key Features

- Fiber-coupled optical input via FC/PC multimode connector
- Single-photon detection based on SPAD technology
- Digital SMA output in LVTTL format (2.5 V, 50 Ω)
- USB Type-C output in LVDS format
- Integrated thermoelectric cooling (TEC)
- Dual power supply options via USB Type-C or external DC input

5 Components Overview

Overview of the key components of the device.

1. 2.1/5.5 mm coaxial DC connector for power supply
2. LED status indicator
3. Digital SMA output
4. 2x dual function USB Type-C ports: can be used as digital outputs (LVDS) or for power supply
5. Fiber-coupled optical input via FC/PC multimode connector





6 Scope of Delivery

- SPAD detector
- Wall-mount DC power converter
- USB Type-C power cable
- SMA coaxial cable
- USB Type-C LVDS signal cable
- FC/PC multimode fiber patch cord
- Documentation

7 Setup

Install the device on a stable laboratory surface. Before installation:

- inspect the device and accessories for visible damage
- verify connectors and cables are intact
- inspect the FC/PC optical input for dust or contamination
- verify that the incident optical power does not exceed the limits given in Table 1

> Do not connect a laser directly to the SPAD.

8 Connections

Power connection

- Connect the device either via USB Type-C power input (5 V DC) or via the 2.1/5.5 mm coaxial DC connector.
- For external powering, use a 9 V DC supply.
- Verify that the status LED illuminates green.

Optical input connection

- Connect a multimode optical fiber with FC/PC connector to the optical input port.
- Ensure proper connector mating and avoid mechanical stress on the fiber.

Digital output connection

- Connect the digital output either via the SMA connector (2.5 V LVTTL, 50 Ω) or via the USB Type-C interface configured for LVDS output.
- Use compatible cables and verify that the connected acquisition system supports the selected signal standard.

9 Configuration

After connecting the device:

- verify that the status LED is illuminated green
- verify that the optical input is correctly connected
- confirm that the selected digital output is connected to a compatible acquisition system

The detector operates autonomously and does not require user-adjustable internal parameters for standard operation.

10 Operation

Once the power supply, optical input, and output connections have been correctly established, the SPAD operates autonomously without further user intervention. Each detected photon event generates a digital output pulse at the SMA output and at the USB Type-C LVDS output.

Normal operation is indicated by a green status LED.

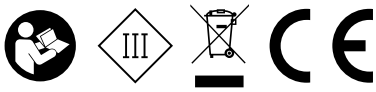
11 Further Information

Detailed information on troubleshooting, maintenance, storage, and disposal is provided in the full operation manual available at https://www.flimlabs.com/products/single-photon-detector/.

12 Product Label

Single Photon SPAD Detector

P/N: SPAD-050-C-X-FC
S/N: SPAD-XXXX-XXX
5 V ⎓ 0.6 A | IP40



MADE IN ITALY - **FLIM LABS S.r.l.**
Via della Farnesina 3 - 00135 Rome
techsupport@flimlabs.com

13 Technical Specifications

Item	SPAD-050-C-B-FC	SPAD-050-C-R-FC
Electrical Specifications		
Electrical safety class	Class III	
Power supply	9 V DC (2.1/5.5 mm) or 5 V DC via USB Type-C	
Maximum current consumption	0.6 A	
Maximum power consumption	3 W	
Signal and Optical Specifications		
Sensor type	Single Photon Avalanche Diode (SPAD)	
Photosensitive area	50 µm	
Spectral response range	370 nm – 900 nm	400 nm – 1000 nm
Peak sensitivity	450 nm	630 nm
Maximum incident optical power	50 µW	
Maximum count rate	3.5 Mcps	3 Mcps
Dark count rate (typical)	7 cps	20 cps
Optical input	1x FC/PC multimode fiber connector	
Outputs	1x SMA (LVTTL) 1x USB Type-C (LVDS)	
Output pulse amplitude	SMA: 2.5 V LVTTL (50 Ω) USB Type-C: LVDS	
Output pulse width	10 ns	
Timing jitter	65 ps RMS	
Impedance	50 Ω	
Physical Specifications		
Dimensions (L x W x H)	127 mm x 77 mm x 40 mm	
Weight	344 g	
Protection rating	IP40	
Environmental Specifications		
Operating temperature	0°C to +40°C	
Storage temperature	-20°C to +70°C	
Relative humidity	Up to 80% non-condensing	

Table 1: Key specifications of the Single Photon SPAD Detector.

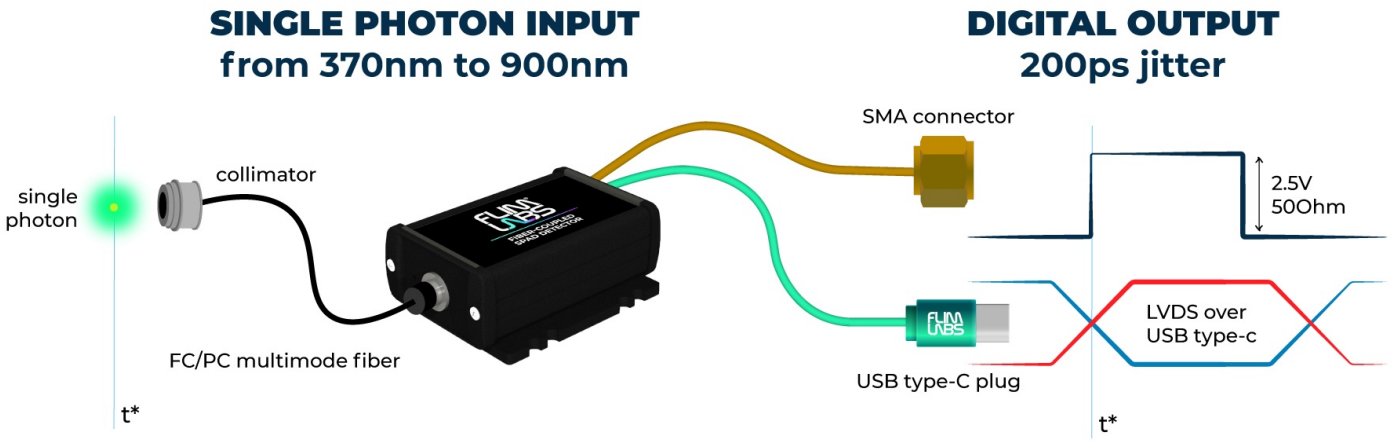


Figure 1: Input and output signal of a SPAD.